7 **(a)** The scale drawing shows the position of a castle, *C*.
The scale is 1 centimetre represents 2 kilometres.

Scale: 1 cm to 2 km

Micah walks at 4.8 km/h for 3 hours on a bearing of 236° from *C*, to his house, *H*.

Mark the position of *H* on the scale drawing.

[3]

(b) A restaurant, *R*, is on a bearing of 083° from *C*.

(i) Work out the bearing of *C* from *R*.

..... [2]

(ii) The distance, d km, from R to C is 5 km, correct to the nearest kilometre.

Complete this statement about the value of d.

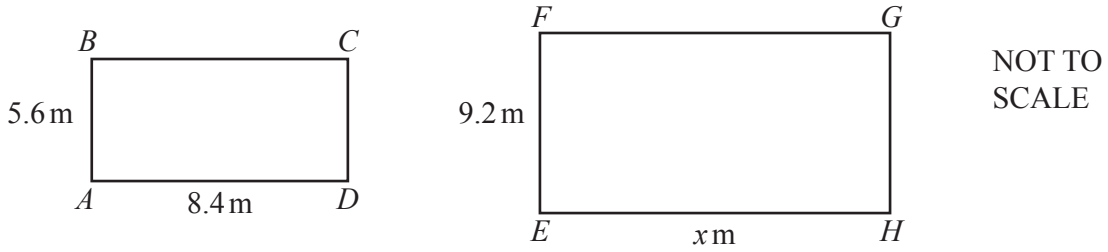
..... $\leqslant d <$ ..... [2]

(c) A model of the castle is made.
On the model, the length of one of the castle walls is 5 centimetres.
The actual length of the wall is 90 metres.

Find the scale of the model in the form $1 : n$.

1 : ..... [2]

(d) The castle has two kitchens with rectangular floors.



Rectangle $ABCD$ is mathematically similar to rectangle $EFGH$.

Calculate the value of x.

$x =$ ..... [2]

(e) The castle has a cylindrical tower.
The tower has a radius of 2.4 m and a height of 25 m.

Calculate the curved surface area of the tower.
Give the units of your answer.

.... [3]